

First steps towards Continual Learning - Ekya

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Ekya: Continuous Learning of Video Analytics Models on Edge Compute Server



- **Authors** : Romil Bhardwaj, Microsoft and UC Berkeley; Zhengxu Xia, University of Chicago; Ganesh Ananthanarayanan, Microsoft; Junchen Jiang, University of Chicago; Yuanchao Shu, Nikolaos Karianakis, Kevin Hsieh, and Paramvir Bahl, Microsoft; Ion Stoica, UC Berkeley
- **Conference** : 19th USENIX Symposium on Networked Systems Design and Implementation



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- **Conference** : 19th USENIX Symposium on Networked Systems Design and Implementation
- **Motivation** : *“Balancing retraining and inference under tight GPU constraints.”*



Need for video analytics



Need for video analytics



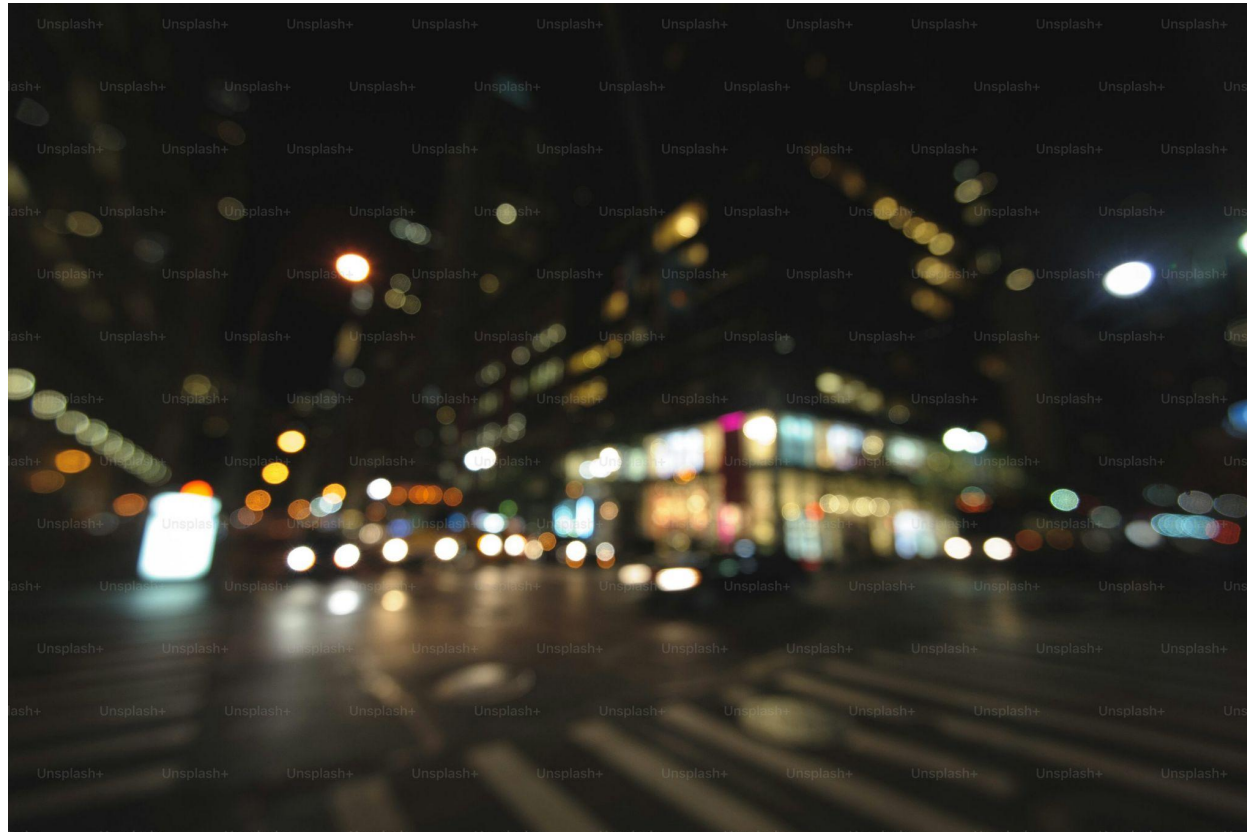
- Video data is everywhere!



Need for video analytics



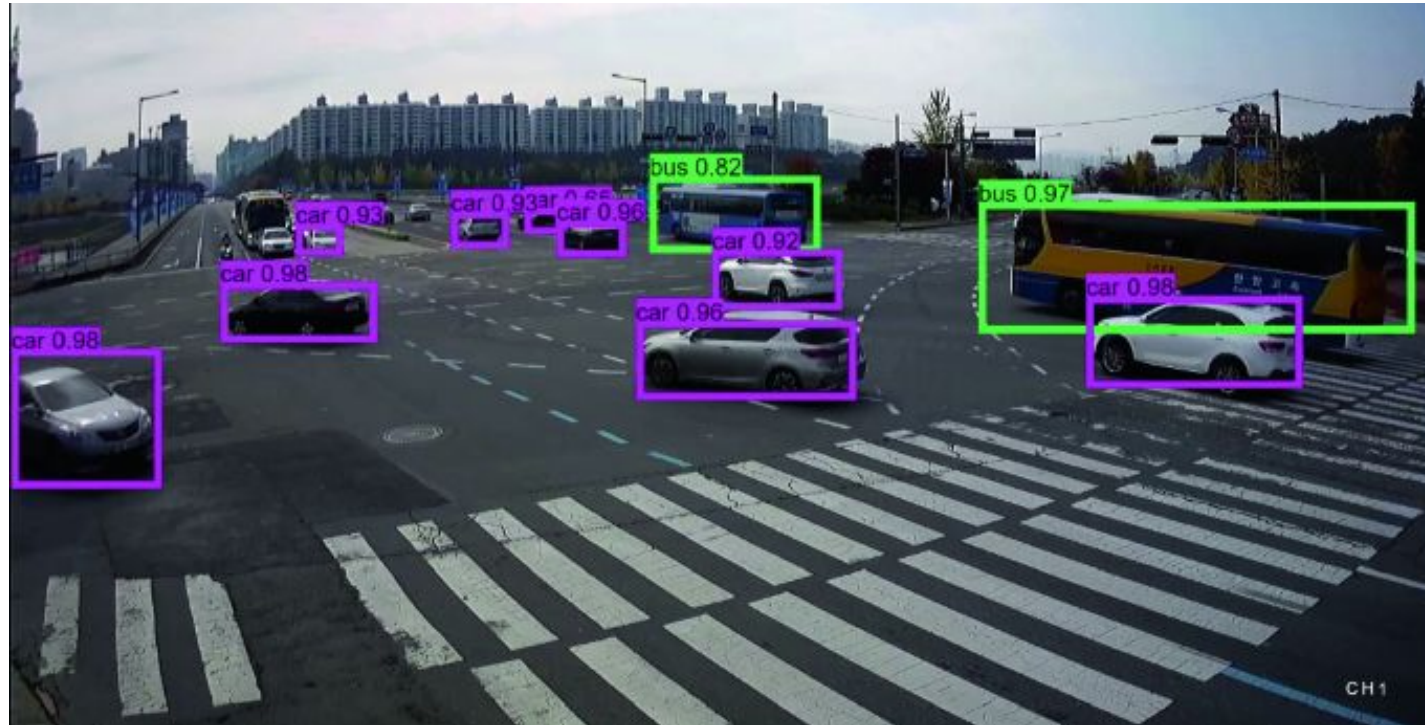
- Video data is everywhere!
- Often, video data is **unstructured**.



Need for video analytics



- Video data is everywhere!
- Often, video data is **unstructured**.
- Requires **ML models** to extract value.



Video analytics at the edge



Video analytics at the edge



1. Uplink cost

- a. Expensive to set up
- b. Not feasible
- c. Unreliable



Video analytics at the edge



1. Uplink cost

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- b. Not feasible
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2. Privacy

- a. Data is sensitive
- b. Regulations



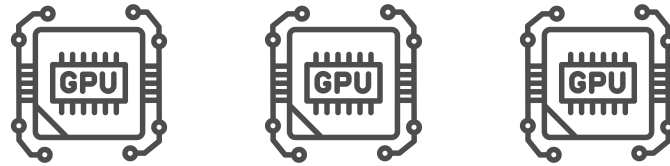
Edge video analytics setup



Edge video analytics setup



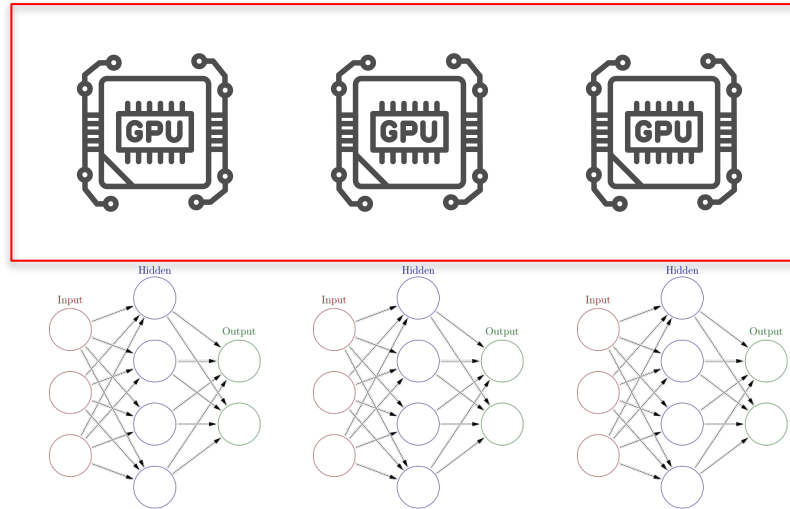
shared resource



Edge video analytics setup



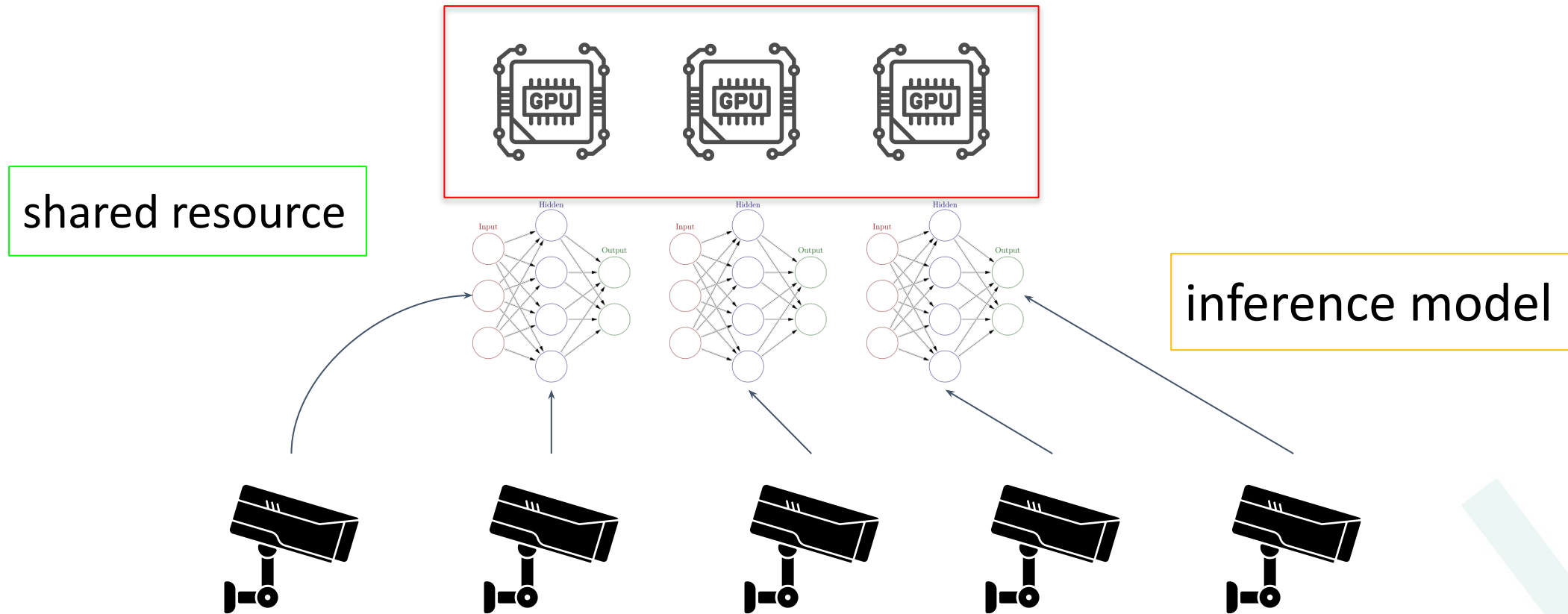
shared resource



inference model



Edge video analytics setup



Challenges with models



Challenges with models



- Limited **generalizability** : smaller models hence limited learnability.



Challenges with models



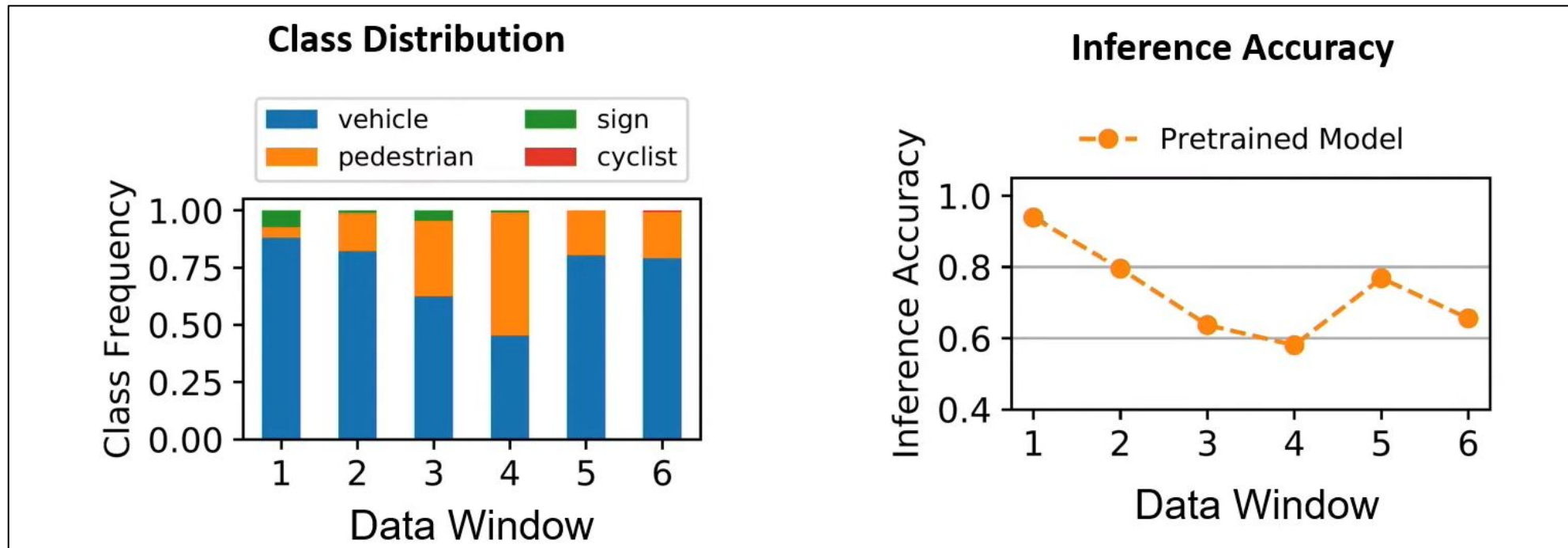
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- **Data drift** : real-world data changes over time , resulting in reduced accuracy of inference model.



Challenges with models



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- **Data drift** : real-world data changes over time , resulting in reduced accuracy of inference model.



Solutions with continuous learning



Solutions with continuous learning



- Adaptive models by **continuously learning** from incoming data.



Solutions with continuous learning



- Adaptive models by **continuously learning** from incoming data
- **Retrain** models periodically.



Solutions with continuous learning



- Adaptive models by **continuously learning** from incoming data
- **Retrain** models periodically.
- **Higher mean accuracy**



Cost to continuous learning



Cost to continuous learning



- Retraining windows require **precious GPU time**.



Cost to continuous learning



- Retraining windows require **precious GPU time**.
- **Allocate resource** for inference and training.



Cost to continuous learning



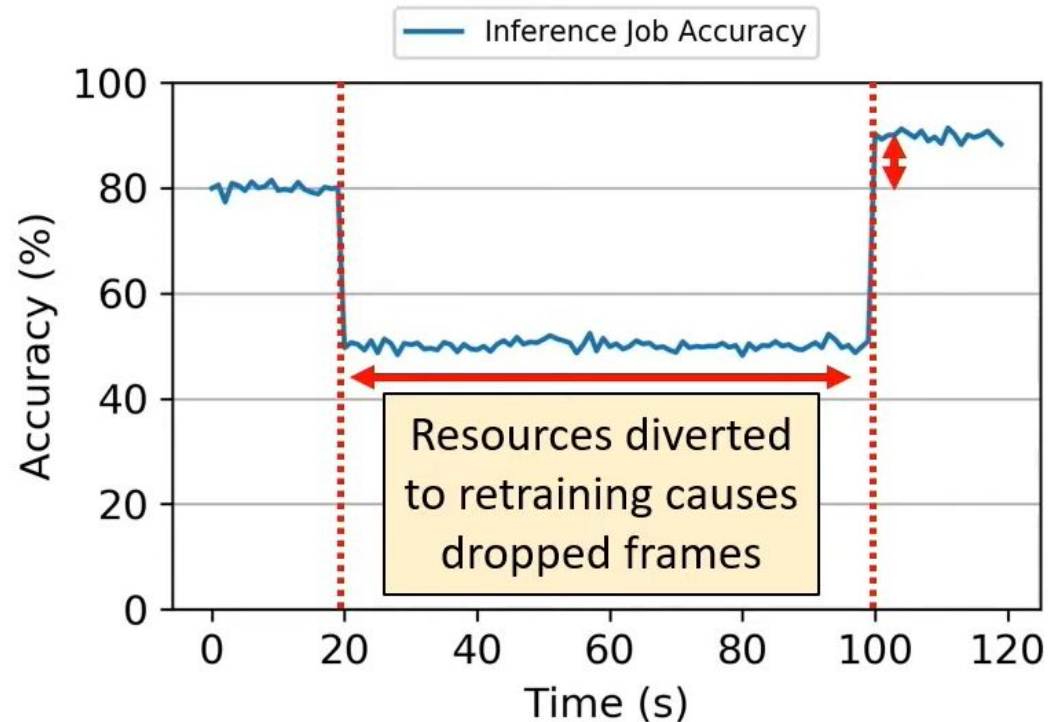
- Retraining windows require **precious GPU time**.
- **Allocate resource** for inference and training.
- Impacts **inference accuracy**.



Cost to continuous learning



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- **Allocate resource** for inference and training.
- Impacts **inference accuracy**.



Resource demands



Resource demands



- Depends on the **configurations/hyperparameters**.



Resource demands



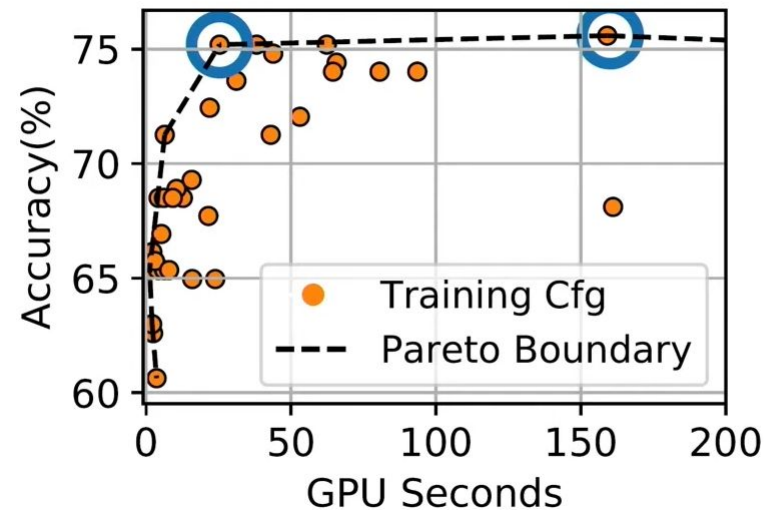
- Depends on the **configurations/hyperparameters**
- **Hyperparameters**
 - Layers to train
 - Sampling rate
 - Number of epochs
 - etc.



Resource demands



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- **Hyperparameters**
 - Layers to train
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 - etc.



Goal



Goal



Objective : **maximize mean accuracy** across all cameras by picking configurations and resource allocations.



Goal



Objective : **maximize mean accuracy** across all cameras by picking configurations and resource allocations.

Subject : **resource capacity constraints**
minimum inference accuracy constraints



Fair scheduler



Fair scheduler



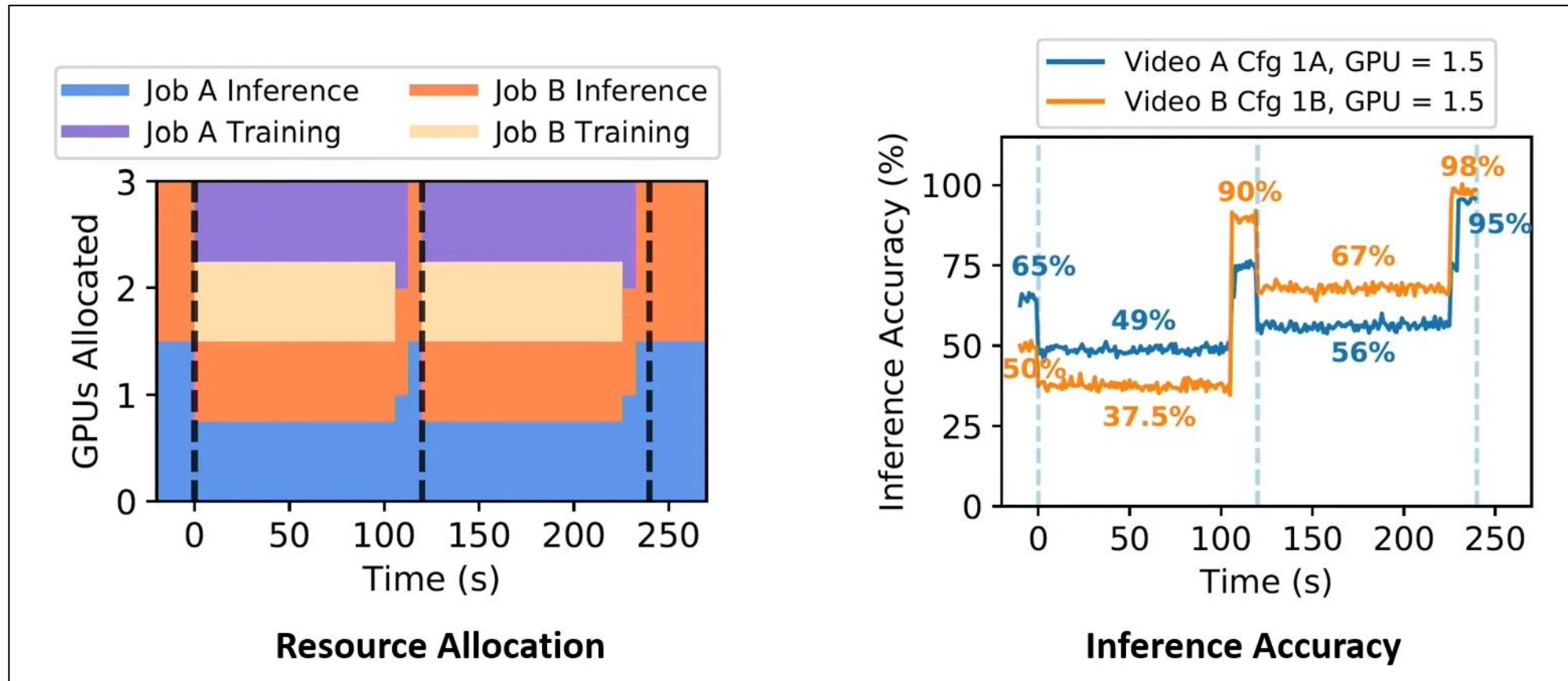
- allocate resources **equally** , pick configurations which give **highest accuracy**.



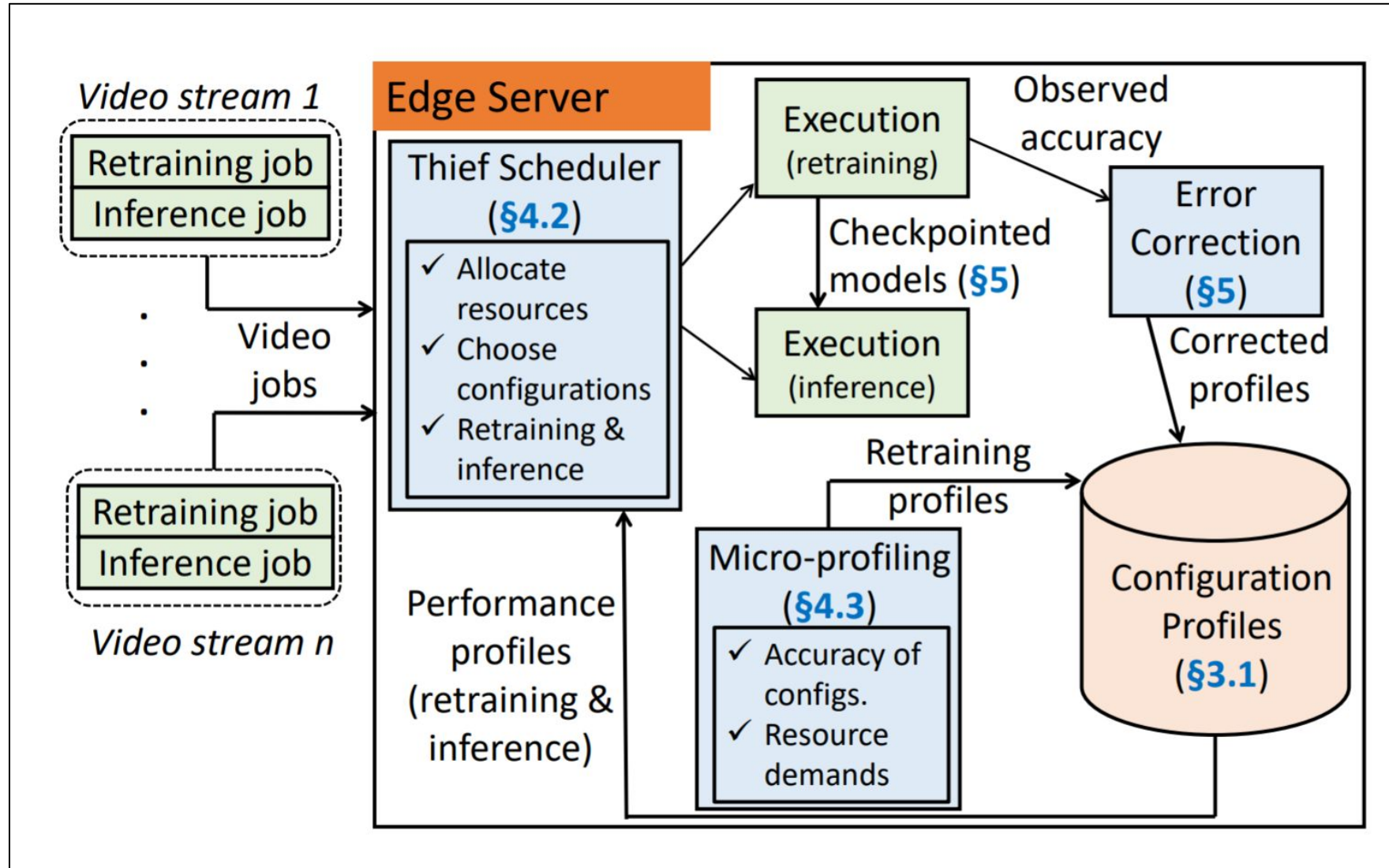
Fair scheduler



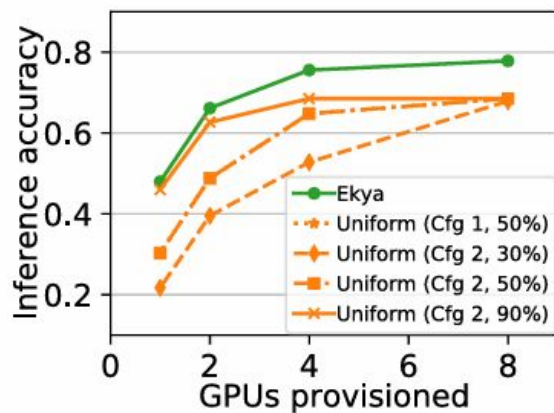
- allocate resources **equally** , pick configurations which give **highest accuracy**



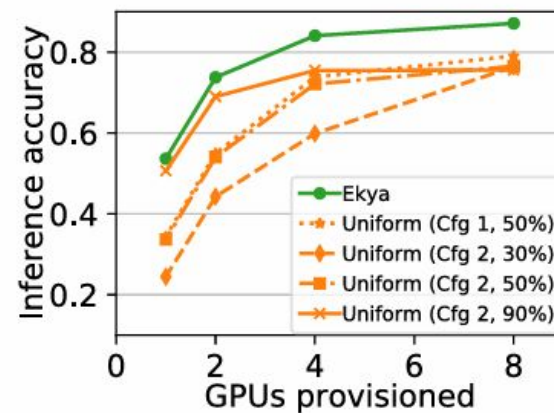
Architecture of Ekya



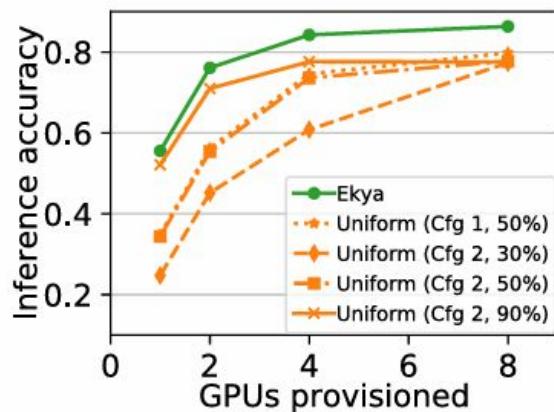
Results



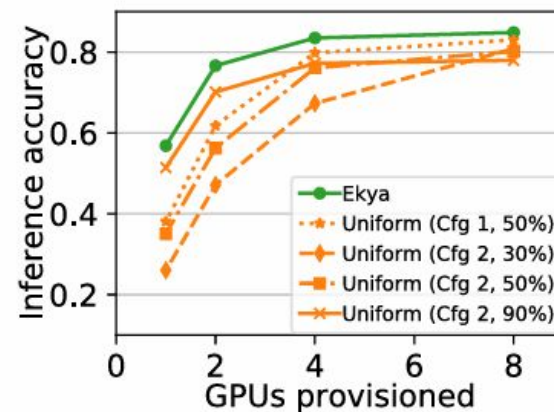
(a) Cityscapes



(b) Waymo



(c) Urban Building



(d) Urban Traffic

Summary



Summary



- **Continual learning** is promising.



Summary



- **Continual learning** is promising.
- **Ekya intelligently allocates resources.**



Summary



- **Continual learning** is promising.
- Ekya **intelligently allocates resources**.
- **Picks hyperparameters efficiently**.



Improvements



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- Early training **may not always** predict accuracy.



Improvements



- Early training **may not always** predict accuracy.
- Thief scheduler is **greedy** (local optimality).



Improvements



- Early training **may not always** predict accuracy.
- Thief scheduler is **greedy** (local optimality).
- **Fixed** retraining windows.



Thank you

A decorative graphic in the bottom right corner consisting of several parallel teal bars of varying lengths, arranged in a diagonal pattern.